LAND SUBSIDENCE SUSCEPTIBILITY MAPPING AND SPATIO-TEMPORAL FORECASTING BY INTEGRATING SENTINEL-1 INSAR IMAGERY WITH MACHINE AND DEEP LEARNING MODELS

A Thesis

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This thesis is dedicated to me

Acknowledgments

The is my Acknowledgments

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Abstract

Land subsidence has become an escalating geohazard in rapidly urbanizing regions such as East Baton Rouge Parish (EBRP), Louisiana, where groundwater extraction, infrastructure loading, and geomorphological variability jointly drive ground deformation. Continuous monitoring and forecasting are therefore critical for infrastructure protection and sustainable water–resource management. This study presents an integrated framework for land–subsidence susceptibility mapping and spatio–temporal forecasting by combining Sentinel–1 Small Baseline Subset Interferometric Synthetic Aperture Radar (SBAS–InSAR) observations with machine–learning and deep–learning models. Multi–temporal SBAS–InSAR data spanning 2017–2025 (246 descending line–of–sight scenes) were analyzed alongside fifteen (15) geospatial conditioning factors, including geology, land–use/land–cover, elevation (DEM), slope, aspect, topographic position index (TPI), curvature, topographic wetness index (TWI), precipitation, and distances to rivers, roads, faults, railways, bridges, and interpolated groundwater well influence.

Gaussian noise–based data augmentation was applied to mitigate data gaps and enhance model generalization. Susceptibility mapping identified the Extra Trees Regressor as the optimal spatial model, achieving a test R^2 of 0.9063 and reduced overfitting relative to Random Forest ($R^2 = 0.8037$). A physics–informed spatio–temporal modeling approach integrating Long Short–Term Memory (LSTM) temporal learning with adaptive stabilization was applied to forecast land subsidence, yielding an overall R^2 of 0.9368 and a Pearson correlation of 0.9723. SHAP analysis highlighted land–use/land–cover, fault proximity, and River–DEM interactions as dominant subsidence drivers. Projections to 2030 indicate gradual deformation stabilization, with a mean velocity of -0.530 mm/year and cumulative

displacement of -6.42 mm.

Chapter 1. Introduction

1.1. Background

Land subsidence is a gradual or sudden downward displacement of the Earth's surface caused by both natural and anthropogenic processes. While natural mechanisms such as sediment consolidation, glacial isostatic adjustment, and tectonic motion contribute to subsidence, human activities are now recognized as the dominant drivers in most severe global cases, accounting for more than 80% of documented high-impact events worldwide (Galloway et al., 1999; Higgins, 2016) . Excessive groundwater extraction, aquifer-system compaction, soil consolidation, and static loading from urban infrastructure significantly increase subsidence rates, posing substantial risks to infrastructure stability, groundwater sustainability, flood resilience, and environmental systems in rapidly urbanizing regions. East Baton Rouge Parish (EBRP), Louisiana, represents a prominent subsidence-prone urban area where geological, hydrological, and anthropogenic factors interact in a complex manner. The region is characterized by low-relief topography and unconsolidated sediments, making it highly sensitive to pore-pressure reduction caused by intensive pumping from the Southern Hills Aquifer System. Rapid urban expansion and heavy infrastructure loading further exacerbate ground deformation. Structural controls such as the Baton Rouge and Denham Springs fault systems introduce spatially heterogeneous and non-linear deformation patterns, increasing vulnerability to flooding, pipeline rupture, and foundation failure (Wang et al., 2021; Abdalla et al., 2024).

Traditional land-subsidence monitoring techniques, including geotechnical surveys, levelling campaigns, and GNSS/GPS measurements, provide high accuracy at discrete locations but are limited by sparse spatial coverage, high operational costs, and insufficient temporal resolution for regional-scale risk assessment. The advent of satellite-based Interferometric Synthetic Aperture Radar (InSAR) has transformed subsidence monitoring by enabling millimetre-level deformation measurements over large areas. In particular, time-series approaches such as Persistent Scatterer InSAR (PS-InSAR) and Small Baseline Subset InSAR (SBAS-InSAR) have proven effective for detecting subtle, long-term deformation in urban and mixed land-cover environments (Ferretti et al., 2001; Berardino et al., 2002; Ferretti et al., 2011b).

Despite these advances, converting large-scale InSAR observations into reliable and interpretable spatio-temporal forecasts remains challenging due to the non-linear, multi-factor nature of subsidence processes. Physically based numerical models offer insight into deformation mechanisms but often require detailed soil and hydrogeological parameters that are rarely available at regional scales (Zhou et al., 2022). Recent developments in machine learning have demonstrated strong potential for addressing these challenges by learning complex relationships between deformation and environmental controls directly from data. In particular, tree-based ensemble models such as the Extra Trees Regressor have shown strong performance in spatial susceptibility mapping due to their robustness to multicollinearity, ability to model non-linear interactions, and reduced overfitting compared to conventional Random Forest approaches (Cieslik et al., 2022; Gao et al., 2024).

While ensemble tree models are well suited for spatial susceptibility analysis, they do not explicitly capture temporal dependencies. Long Short-Term Memory (LSTM) networks address this limitation by modelling sequential deformation dynamics and long-term temporal dependencies within InSAR time series. However, purely data-driven LSTM models may generate physically unrealistic trends when extrapolated beyond the observation period. To overcome this limitation, recent studies have emphasized the integration of deep learning with physics-based constraints to improve forecast stability and geomechanical plausibility (Zhu et al., 2024).

Guided by this framework, the present study integrates SBAS-InSAR–derived deformation, Extra Trees–based spatial susceptibility mapping, and a physics-informed LSTM spatio-temporal forecasting approach enhanced with explainable artificial intelligence. This integrated methodology provides a robust, interpretable, and geomechanically consistent framework for assessing present-day subsidence patterns and generating reliable projections of ground deformation in East Baton Rouge Parish through 2030.

1.2. Thesis Objectives

The main research objectives are to:

i. Generate high-resolution land-subsidence map for East Baton Rouge Parish using Sentinel-1 SBAS-InSAR data (2017–2025)

ii. Identify and rank key environmental and anthropogenic drivers for land subsidence using EXtra Trees and Random Forest machine-learning model with SHAP-based Explainability.

iii. Develop a physics-informed LSTM-based spatio-temporal model to forcast land subsidence from InSAR time-series data, ensuring geomechanically consistent projections through 2030.

1.3. Thesis Organization

This thesis is structured into six chapters as follows:

Chapter 1, this chapter, introduces the research topic of land subsidence susceptibility mapping and spatio-temporal forecasting. It provides background information, outlines research objectives, and presents the thesis organization.

Chapter 2 presents the study area characteristics of East Baton Rouge Parish, including its geological setting, hydrogeological conditions, and historical subsidence patterns that make it vulnerable to ground deformation.

Chapter 3 provides a comprehensive literature review covering: advances in InSAR monitoring techniques for subsidence, machine learning applications in susceptibility mapping, deep learning approaches for time-series forecasting, hybrid modeling frameworks, and methods for analyzing subsidence driving factors. The chapter concludes by identifying research gaps this study aims to address.

Chapter 4 details the methodology employed, including: PS-InSAR processing of Sentinel-1 data, preparation of environmental and anthropogenic predictor variables, development of machine learning models (Random Forest) for susceptibility mapping, implementation of deep learning models (LSTM, MLP, SVR) for temporal forecasting, and construction of the hybrid stacked ensemble framework. The chapter also describes validation procedures using GNSS data and SHAP-based interpretation methods.

Chapter 5 presents the analysis results and discussion, covering: spatial patterns of observed subsidence, relative importance of driving factors, performance evaluation of individual and ensemble models, and validation against ground-truth measurements. The implications for urban planning and groundwater management are also discussed.

Chapter 6 summarizes key conclusions, provides recommendations for land subsidence monitoring and mitigation in East Baton Rouge, acknowledges study limitations, and suggests directions for future research.

Chapter 2. Study Area and Materials Used

2.1. Study Area

This study is conducted in East Baton Rouge Parish (EBRP), located in southeastern Louisiana, United States, within the Baton Rouge Capital Area. The parish covers approximately 471 square miles and extends between 30°18'–30°36'N latitude and 91°06'–90°48'W longitude (Figure 2.1). The study area is bounded by major hydrological systems, including the Mississippi River to the west and the Comite River to the east. It is characterized by low-relief floodplains, forested wetlands, and extensive alluvial deposits typical of the Gulf Coast region. This geographic setting places East Baton Rouge Parish within a broader zone of documented subsidence and aquifer-system response associated with sedimentary basins in the southern United States (Galloway et al., 2000).

Geologically, East Baton Rouge Parish is underlain by thick sequences of unconsolidated Holocene and Pleistocene deltaic and fluvial sediments, composed primarily of clay, silt, and fine sand. These young, poorly consolidated materials exhibit high compressibility and low shear strength, making the region particularly susceptible to land subsidence and soil deformation when subjected to changes in effective stress (Higgins, 2016). Such geological conditions are characteristic of sedimentary basin and coastal plain environments, where compaction-driven subsidence is commonly observed.

Land subsidence in EBRP is driven predominantly by anthropogenic activities, especially long-term groundwater extraction from the Baton Rouge aquifer system. Sustained withdrawal of groundwater reduces pore pressure within aquifer systems, leading to aquifer-system compaction and progressive lowering of the land surface (Galloway et al., 2000). These effects may be irreversible where inelastic compaction occurs. Subsidence associated with groundwater depletion has been widely documented across sedimentary basins in the United States and is increasingly mapped using geodetic techniques (Higgins, 2016).

In addition to groundwater withdrawal, urban development and surface loading contribute to subsidence processes within East Baton Rouge Parish. Expansion of infrastructure and increased building density impose additional loads on compressible sediments, accelerating consolidation in low-lying areas. The combined effects of fluid withdrawal and surface loading produce complex deformation patterns that vary spatially and temporally, emphasizing the need for spatially continuous monitoring approaches such as Interferometric Synthetic Aperture Radar (InSAR) (Ferretti et al., 2001; Crosetto et al., 2016).

Climatic and hydrological conditions further influence ground deformation in the study area. East Baton Rouge Parish experiences a humid subtropical climate, with high annual precipitation, intense rainfall events, and periodic flooding. Seasonal variations in groundwater levels, soil moisture, and surface water storage can induce transient and nonlinear deformation signals, which may be superimposed on long-term subsidence trends. These effects complicate the interpretation of InSAR time series and require careful consideration of atmospheric and hydrologic contributions to observed displacement (Berardino et al., 2002; Fattahi and Amelung, 2015).

Given the combined influence of compressible deltaic sediments, intensive groundwater extraction, urban loading, and hydro-climatic variability, East Baton Rouge Parish represents an ideal natural laboratory for investigating land subsidence using Sentinel-1 time-series InSAR. The spatial extent of the study area supports high-resolution deformation mapping and facilitates integration with GNSS control points for validation and uncertainty assessment.

Improved understanding of the spatial and temporal characteristics of subsidence in EBRP is essential for sustainable groundwater management, infrastructure resilience, and long-term planning in sedimentary basin environments (Crosetto et al., 2016; Galloway et al., 1999).

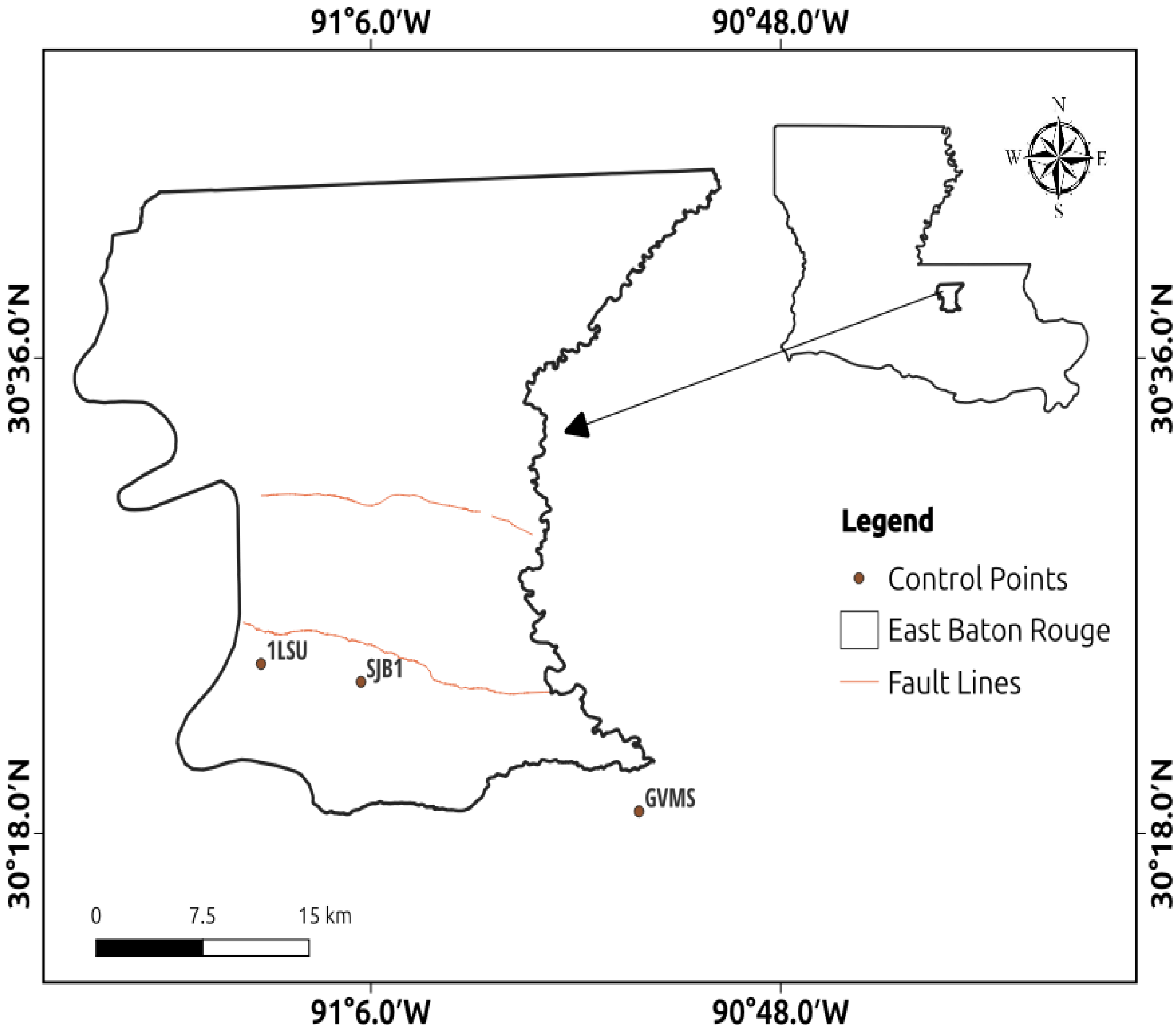

Figure 2.1. Location map of East Baton Rouge Parish

2.2. Materials Used

This study utilises a comprehensive set of satellite data, auxiliary geospatial datasets, atmospheric products, and open-source software tools to drive ten (8) years (2017-2025) of ground deformation and velocity estimates using timeseries SBAS-InSAR techniques. The materials used are described in detail below.

2.2.1. Sentinel-1 SAR Dataset

A total of 246 Sentinel-1A descending line-of-sight (LOS) Single-Look Complex (SLC) scenes were used in this study. The data were acquired in Interferometric Wide (IW) mode with VV polarisation in descending tracks. Sentinel-1 is a C-band (5.405 GHz) SAR mission operated by the European Space Agency (ESA) and provides systematic, long-term observations suitable for deformation monitoring.

2.2.2. Precise Orbit Ephemerides

Precise Orbits Ephemerides (POEORB) provided by the ESA were applied to all Sentinel-1 acquisitions prior to interferometric processing. Compared to restituted orbits, POEORB significantly improves satellite position accuracy, reducing residual orbital errors and mitigating long-wavelength phase ramps in interferograms (Ferretti et al., 2011a).

2.2.3. Digital Elevation Model and Atmospheric Data

A 30 m resolution Digital Elevation Model (DEM) was used to correct for topographic phase contributions in interferograms generation. The DEM also served as the basis for deriving topographic conditioning factors used in the susceptibility analysis.

To mitigate atmospheric artefacts, particularly stratified tropospheric delays, ERA5 (Fifth Generation ECMWF Atmospheric Reanalysis) data were incorporated during timeseries processing. ERA5 provides high-resolution global estimates of atmospheric state variables and has been widely adopted for atmospheric phase screen correction in InSAR studies (Hersbach et al., 2020).

2.2.4. Geospatial Conditioning Factors

A total of fifteen (15) geospatial conditioning factors were prepared at 30 m resolution and aligned to a common spatial grid for land subsidence susceptibility modelling. These factors were grouped into topographic, geological, hydrological, structural/tectonic, infrastructure, climatic, and anthropogenic categories, reflecting the multi-causal nature of land subsidence processes.

- Topographic factors: Digital Elevation Model (DEM), slope, aspect, Topographic Position Index (TPI), curvature, Topographic Wetness Index (TWI).
- Geological factors: geology, Land Use/Land Cover (LULC).
- Hydrological factors: distance from river.
- Structural/Tectonic factors: distance from fault (Baton Rouge and Denham Springs fault systems).
- Infrastructure factors: distance from road, distance from railway, distance from bridge.
- Climatic factors: precipitation.
- Anthropogenic factors: well_idw (interpolated groundwater well influence).

Continuous layers were min–max scaled; categorical layers (geology, LULC) were one-hot encoded.

2.2.5. Validation Data

InSAR-derived deformation was validated using GNSS vertical displacement time series from the GulfNet CORS stations (1LSU and GMVS) in EBRP obtained from the Nevada Geodetic Laboratory (NGL). GNSS provides high-precision point-based measurements and serves as a standard benchmark for InSAR timeseries validation (Hanssen, 2001; **?**)

Chapter 3. Literature Review

3.1. Overview of Land Subsidence in Urban Areas

Land subsidence is increasingly recognized as a major global geohazard, with over 80% of severe cases associated with aquifer compaction and groundwater withdrawal (Galloway et al., 1999). In rapidly urbanizing regions like the Lower Mississippi Basin, the interplay between shallow geology and heavy infrastructure growth intensifies surface deformation. Urban environments are particularly susceptible due to the high density of man-made structures that act as static loads on compressible soil layers. Research indicates that unmitigated subsidence triggers a cascade of secondary hazards, including pipeline ruptures, building tilting, and structural failures, resulting in significant economic losses (Wang et al., 2021).

Land subsidence, which is defined as the slow downward movement of the Earth's surface, is becoming more widely acknowledged as a significant geohazard in areas of the world that are rapidly becoming more urbanized (Galloway et al., 2023). In urban environments, subsidence processes are fueled by excessive groundwater extraction, sediment consolidation, urban infrastructure loading, and land-use change (Holzer and Johnson, 1985). Research on the Lower Mississippi Basin, for instance, demonstrates how shallow geology and infrastructure growth can intensify surface deformation and compaction (Wang et al., 2021). The hazards include diminished drainage capacity, heightened susceptibility to flooding in low-lying urban corridors, and harm to pipelines and foundations. Subsurface faulting and structural geology make subsidence patterns in many cities even more complex (Younas et al., 2023). These findings highlight the need for reliable frameworks that can both map the areas where subsidence is most likely to occur and predict its future course.

3.2. Advancements in InSAR Monitoring for Subsidence

The development of Synthetic Aperture Radar Interferometry (InSAR) has transformed ground-deformation monitoring by providing high-resolution, wide-area observations. Traditional geodetic techniques, such as GNSS and levelling, have historically provided precise point measurements but are limited by prohibitive costs and low spatial resolution.

3.2.1. Methodological Evolution

Early Differential InSAR (D-InSAR) was limited by atmospheric delay and spatiotemporal decorrelation. Multi-temporal approaches like Persistent Scatterer InSAR (PS-InSAR) improved accuracy by identifying stable radar targets that retain high phase stability over time, achieving sub-millimeter precision on built infrastructure (Ferretti et al., 2001, 2011b). However, PS-InSAR is often sparse in non-urbanized or vegetated areas, limiting its applicability to mixed land-cover environments.

Small Baseline Subset (SBAS-InSAR) has emerged as a superior alternative, particularly in regions like Louisiana with mixed urban and vegetated terrain (Berardino et al., 2002). By utilizing multiple master interferograms with small perpendicular baselines, SBAS identifies Slowly Decorrelating Filtered Pixels (SDFP), which significantly reduces the impact of spatial and temporal decorrelation. Recent studies indicate that processing high-volume datasets—such as the 246 descending Sentinel-1 scenes (2017–2025) used in this research—allows for the detection of both linear and complex non-linear deformation trends with millimetre-level accuracy. Comparatively, SBAS-InSAR provides up to 2.83 times higher monitoring density than PS-InSAR in mixed urban-rural environments (Bayik et al., 2025).

Unlike traditional geodetic methods, InSAR enables millimeter-scale motion detection across vast regions regardless of weather or lighting conditions (Ferretti et al., 2020; Crosetto et al., 2016). Building on these advances, newer methods such as Distributed Scatterer (DS-InSAR) and Multi-Temporal InSAR (MT-InSAR) extend coherence to mixed urban–rural environments by statistically analyzing pixel clusters (Fattahi and Amelung, 2015).

3.2.2. Atmospheric Correction and Data Quality

Modern InSAR frameworks integrate the Generic Atmospheric Correction Online Service (GACOS) to mitigate tropospheric delay, significantly reducing the root-mean-square (RMS) error of phase measurements. This enhancement is critical for long-term monitoring in tropical and subtropical regions where water vapor fluctuations are substantial. The advent of open-access Sentinel-1A/B data with a 12-day revisit cycle has further enhanced InSAR's operational value for regional subsidence monitoring.

Numerous studies have demonstrated the effectiveness of SBAS-InSAR: Hakim et al. (2023) used PS-InSAR coupled with hybrid CNN–SVR models to improve deformation prediction in Indonesia; Li et al. (2020) applied PS-InSAR and LSTM networks in Beijing, achieving $R^2 > 0.85$ between deformation and aquifer changes; and Yu et al. (2023) integrated PS-InSAR with GNSS in Houston to map displacements of 5–15 mm yr^{-1}. More recent works, including Peng et al. (2024) and Liu et al. (2023), show that data-driven fusion of InSAR with deep-learning architectures like LSTM can outperform traditional regression models in temporal subsidence forecasting.

Beyond urban analysis, InSAR has advanced environmental and hydrogeological applications, linking surface deformation to aquifer compaction, groundwater extraction, and soil-structure interactions. For instance, Wang et al. (2023) used InSAR and RF regression to identify groundwater depth as the main driver of subsidence in China, while Li et al. (2020) demonstrated strong correlations between deformation and groundwater-level fluctuations. Overall, InSAR's ability to deliver dense, long-term deformation time series combined with its compatibility with machine-learning and deep-learning frameworks makes it an indispensable tool for both monitoring and forecasting. Yet, most existing studies emphasize *where* deformation occurs rather than *how and when* it evolves. The present research bridges this gap by integrating SBAS-InSAR time-series data with hybrid deep-learning models to perform spatio-temporal forecasting and assess environmental causative factors driving subsidence across East Baton Rouge.

3.3. Machine Learning for Subsidence Susceptibility Mapping

Machine learning (ML) has become the global standard for modelling the non-linear relationship between ground deformation and various geospatial drivers, such as Land Use/Land Cover (LULC), soil types, and distance from faults. Gharechaee et al. (2023) applied RF, KNN, and CART to InSAR-derived subsidence in Iran's Bakhtegan Basin, finding RF performed best and highlighting that the leading susceptibility factors were not only groundwater withdrawal but also proximity to dams, faults and mines. Orlandi et al. (2024) compared ML approaches in Murcia, Spain, using land-use, lithology, and other predictors, demonstrating the utility of ML for zonation and factor ranking.

3.3.1. Ensemble Learners and Extra Trees Regression

While Random Forest (RF) remains a common benchmark due to its ability to handle multi-variate data without pre-defined assumptions, it can be prone to overfitting in complex geological datasets (Breiman, 2001). This study identifies the Extra Trees (ERT) Regressor as a more robust alternative for subsidence susceptibility mapping. By implementing random splits at the node level, ERT enhances model generalisation and reduces variance compared to conventional bagged ensembles (Gao et al., 2024; Mame et al., 2025). In the East Baton Rouge Parish region, Extra Trees achieved a test R^2 of 0.9063, significantly outperforming Random Forest ($R^2 = 0.8037$), demonstrating superior predictive power and reduced overfitting for complex, multi-factor subsidence systems.

3.3.2. Gaussian Process Regression and Data Augmentation

Gaussian Process Regression (GPR) is increasingly used for its ability to model calibrated predictive uncertainty, making it highly effective for small datasets and areas with limited observability. To address the technical challenge of small dataset sizes in local subsidence surveys, Gaussian noise-based data augmentation has been validated as an effective strategy to expand training samples and ensure model stability (Mame et al., 2025). This augmentation approach mitigated data gaps and enhanced model generalization, expanding the effective training dataset to over 390,000 samples. Hosseinzadeh et al. (2024) evaluated seven ML methods (CART, BRT, SVM, RF, logistic regression, etc.) in Iran and found BRT outperformed others in susceptibility mapping. These studies demonstrate ML's strength in spatial patterning and factor ranking, yet many stop short of integrating temporal forecasting or sequence modelling, limiting their value for prediction of future subsidence trends.

3.4. Deep Learning and Time Series Forecasting of Subsidence

Deep-learning (DL) models, especially sequence models like LSTM, are increasingly adopted for subsidence forecasting because of their ability to capture temporal dependencies and non-linear dynamics. Li et al. (2020) applied LSTM to PS-InSAR data in Beijing Plain, showing promise for temporal modelling though with limitations in spatial generalization. Mirmazloumi et al. (2023) used LSTM on InSAR time-series for three mining sites, demonstrating an early-warning capability by forecasting 3–5 steps ahead.

3.4.1. LSTM and Bidirectional Architectures

Long Short-Term Memory (LSTM) networks are adept at capturing non-linear trends in InSAR time series through a memory cell regulated by three gates: the input, forget, and output gates (Hochreiter and Schmidhuber, 1997; Greff et al., 2017). Bidirectional LSTM (BiLSTM) further improves accuracy by capturing information in both forward and backward temporal directions, providing a more holistic understanding of deformation dynamics (Raissi et al., 2019).

Liu et al. (2023) implemented SBAS-InSAR plus an attention-LSTM (AT-LSTM) model for mining subsidence, highlighting how attention mechanisms improve temporal modelling. Soni et al. (2025) developed a modified LSTM for PS-InSAR in mining-induced subsidence, achieving high accuracy ($R^2 \sim 0.98$). These works confirm that DL sequence modelling is powerful for forecasting, but many studies remain site-specific, focus largely on mining rather than urban settings, and rarely integrate causative-factor analysis or ML-based spatial mapping.

3.4.2. CNN-LSTM Hybrids and Attention Mechanisms

Combining Convolutional Neural Networks (CNN) for spatial feature extraction with LSTM for temporal learning significantly improves prediction accuracy for complex non-linear deformations. Lu et al. (2024) developed DACLnet, a Dual-Attention-Mechanism

CNN-LSTM Network for accurate prediction of nonlinear InSAR deformation, achieving state-of-the-art performance in capturing spatial-temporal patterns. The integration of Attention Mechanisms allows models to focus on critical time points, such as sudden shifts in deformation rates caused by seasonal rainfall or peak pumping, thereby improving both accuracy and interpretability.

3.5. Integrated Hybrid Frameworks and Physics Constraints

For spatio-temporal forecasting, the literature increasingly focuses on Long Short-Term Memory (LSTM) networks due to their ability to capture long-term temporal dependencies through gated memory cells. However, standalone data-driven models may lack geomechanical consistency and produce physically unrealistic trends when extrapolated beyond the observation period (Zhu et al., 2024). To resolve this limitation, current research favours hybrid deep-learning frameworks that combine the temporal strengths of LSTM with the nonlinear interaction mapping of Multi-Layer Perceptrons (MLP) and the smooth kernel-based regression of Support Vector Regression (SVR). These components are fused via a stacking ensemble using a meta-learner (MLP), which achieves superior accuracy by leveraging the complementary strengths of each base algorithm.

3.5.1. Stacking Ensembles and Meta-Learning

The convergence of ML (for spatial susceptibility) and DL (for temporal forecasting) into hybrid frameworks is still emerging in the subsidence domain. Stacking multiple base learners (e.g., LSTM, MLP, and SVR) via a meta-learner improves robustness and predictive reliability compared to standalone models. Zheng et al. (2024) compared SVR, Holt smoothing and MLP on InSAR time-series and found MLP versatile but did not extend to LSTM or hybrid stacks. Naghibi et al. (2022) developed an integrated InSAR-machine learning approach for ground deformation rate modelling in arid areas, demonstrating the value of combining remote sensing with ensemble methods. Recent advances by Zhu et al. (2024) employed SBAS-InSAR and CNN-BiGRU-attention models for high-precision subsidence prediction, achieving remarkable accuracy in mining areas and validating the potential of attention-based hybrid architectures.

This study's hybrid stacked ensemble framework achieves an overall R^2 of 0.9368, with a Pearson correlation of 0.9723, providing a physically plausible projection of stabilization through 2030. The framework operates analogously to a medical diagnostic panel: the LSTM serves as the cardiologist examining deformation history over time; the MLP functions as the geneticist identifying how different factors interact non-linearly; and the SVR acts as the radiologist detecting smooth, consistent patterns in the data. A meta-learner (Lead Consultant) optimally weights their predictions to provide the most reliable forecast, ensuring geomechanical plausibility and scientific consistency in the 2030 subsidence projections.

3.5.2. Physics-Informed Machine Learning (PIML)

Purely data-driven models may produce physically implausible results, such as unrealistic acceleration or velocity reversals. Physics-Informed Machine Learning (PIML) frameworks integrate geophysical constraints (e.g., velocity continuity and conservation laws) to ensure that long-term projections remain geomechanically consistent. Raissi et al. (2019) developed Physics-Informed Neural Networks (PINNs), which encode physical laws as penalty terms in the loss function, thereby constraining the model to produce physically realistic predictions. Such gaps suggest the potential for stacked models using meta-learning ensembles to combine base-model predictions and latent features for enhanced accuracy and interpretability. Hybrid models are common in other geohazard domains (e.g., landslides, groundwater) but remain under-utilized in urban subsidence. Thus, implementing a hybrid DL/ML ensemble to both *map* and *forecast* subsidence fills a methodological gap in current literature.

3.6. Subsidence Driving Factor Analysis and Explainability

A critical limitation of traditional deep learning is its "black-box" nature, which hinders its use in urban planning and policy-making. Explainable AI (XAI) tools, specifically SHapley Additive exPlanations (SHAP), are now applied to quantify the marginal contribution of each input factor to the subsidence signal. Liu et al. (2025) applied XGBoost and SHAP on PS-InSAR deformation along the Beijing–Xiong 'an railway, showing groundwater fluctuations contributed approximately 35% of subsidence signal. Xue et al. (2024) used ML on InSAR data to inspect deformation along high-speed rail corridors, emphasizing hydrological and structural controls.

Understanding *why* subsidence occurs and identifying and quantifying drivers is vital for decision-making in resource management and infrastructure planning. SHAP analysis has revealed that while LULC and fault proximity are primary contributors, River-DEM interactions and groundwater fluctuations account for significant portions of the deformation signal in regional contexts. By integrating SHAP within a hybrid forecasting pipeline, transparent and scientifically defensible explanations of subsidence mechanisms can be provided, supporting sustainable water resource and infrastructure management (Chen and Li, 2023). These studies show how driver-ranking can support mitigation strategies and inform policy decisions. Nevertheless, few investigations combine hybrid DL forecasting with explicit causative-factor analysis in urban contexts; the use of SHAP in a hybrid DL/ML framework addresses a real literature gap.

3.7. Data Scarcity and Augmentation Techniques

In geohazard research, obtaining large datasets for model training is often time-consuming and expensive. Gaussian noise-based data augmentation is a validated strategy to improve model stability and generalisation. Adding small amounts of noise allows the training set to expand significantly without distorting the original data distribution, thereby preventing overfitting in tree-based and deep-learning architectures. This approach has been successfully applied in the context of subsidence prediction with Extra Trees models, where synthetic data generation improved model robustness without compromising prediction accuracy (Mame et al., 2025).

3.8. Research Gaps and Study Objectives

While many studies have applied InSAR and machine or deep learning (ML/DL) methods to analyze land subsidence, most have treated spatial susceptibility mapping and temporal forecasting as separate tasks, often overlooking model interpretability (Gharechaee et al., 2023; Rahmani et al., 2024; Fu et al., 2023). Applications within complex U.S. Gulf Coast urban environments also remain limited. Recent research by Qiao (2023) highlighted the specific need for integrated monitoring and assessment frameworks in coastal subsidence-prone regions. To address these gaps, this study integrates Small Baseline Subset InSAR (SBAS-InSAR) time-series data with a hybrid ML/DL ensemble combining Extra Trees for spatial susceptibility mapping and an LSTM-MLP-SVR stacking framework for temporal forecasting, coupled with SHAP (SHapley Additive exPlanations) interpretation to develop predictive and interpretable subsidence models tailored to Baton Rouge's unique environmental and infrastructural conditions (Abdalla et al., 2024; Mojtahedi et al., 2025).

Chapter 4. Methods

4.1. Methodologies

4.1.1. SBAS-InSAR Processing Pipeline

Small Baseline Subset (SBAS-InSAR) was applied to the 246-scene SLC stack to extract deformation time series. SBAS settings prioritised small perpendicular and temporal baselines to identify Slowly Decorrelating Filtered Pixels (SDFP) and ensure dense coverage across mixed urban–vegetated terrain. Atmospheric delays were mitigated with GACOS corrections. Phase unwrapping used the Minimum Cost Flow (MCF) algorithm and deformation time series were geocoded to the study grid.

4.1.2. Gaussian Noise-Based Data Augmentation

To mitigate limited labeled samples, Gaussian noise augmentation was applied by perturbing feature vectors with zero-mean Gaussian noise set to 3% of each parameter's range. The augmentation expanded the training dataset by a factor of 2.5 to a total of 393,412 samples while preserving overall feature distributions.

4.1.3. Spatial Susceptibility Modelling: Extra Trees Regressor

Spatial susceptibility mapping employed the Extra Trees (Extremely Randomized Trees) regressor. Key points:

- Randomised node splits and feature sampling reduce variance and overfitting compared to RF.
- Hyperparameters were tuned via Bayesian optimisation (Tree-structured Parzen Estimator).
- Model evaluation used R^2, RMSE and k-fold cross-validation; feature importance was interpreted with SHAP.

4.1.4. Spatio-Temporal Forecasting: Hybrid LSTM–Physics Framework

A hybrid forecasting pipeline combines data-driven temporal learning with physics-inspired stabilisation:

1. Temporal learner: stacked LSTM (64 then 32 units) trained on 12-step PS time sequences with associated feature groups; Adam optimiser and early stopping employed.

2. Physics-based smoothing: a hyperbolic-tangent filter constrains instantaneous accelerations to a physical threshold (0.35 mm yr^{-2}) to enforce year-to-year velocity continuity.

3. Adaptive stabilisation: an exponential decay term models long-term geomechanical relaxation and prevents unrealistic linear acceleration in multi-year forecasts to 2030.

Base-model out-of-fold predictions (LSTM, MLP, SVR) were combined in a stacking ensemble with an MLP meta-learner to improve forecast robustness.

4.1.5. Explainable AI: SHAP Analysis

SHAP (SHapley Additive exPlanations) was applied to the Extra Trees spatial model and the stacked ensemble outputs to quantify marginal feature contributions. SHAP-derived rankings informed interpretation of dominant drivers (e.g., LULC, fault proximity, River–DEM interactions).

Analogy: SBAS-InSAR acts as the radar sweep; the Extra Trees model provides the hazard map; the Hybrid LSTM–Physics framework is the autopilot where LSTM learns history and physics constraints enforce safe, physically plausible forecasts through 2030.

Chapter 5. References

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